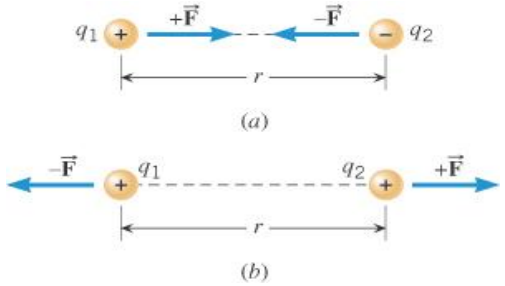
Chapter 35 The Electrostatic Force
Calculus Physics II

Coulomb's Law of Electrostatics – Electrostatic Force - Suppose that two-point charges, with net charges of q_1 and q_2 are a distance r apart in a vacuum. If q_1 and q_2 have the same type of charge, the two charges repel each other; if they have opposite types of charge, they attract each other. The force experienced by either charge due to the other is called a Coulomb or electric force.



- ❖ The forces on the two objects with a net charge are equal in magnitude and opposite in direction even if the charges are not the same (Newton's 3rd law).
- ❖ For two objects with a net charge, the force acts along a straight line drawn through both charges.

The Electric Force is a vector quantity; there is a magnitude and direction!

Point Charge – is a charged object that has a net charge but is very small so that it can be considered a point (at its center) with no dimension to the object and is only a point in space.

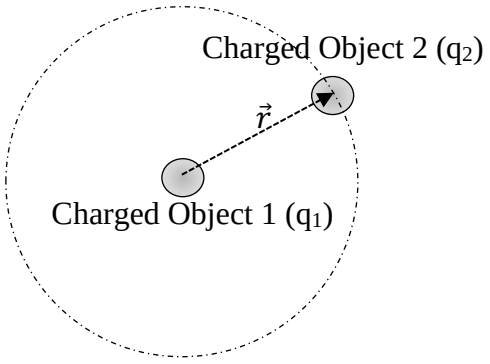


Coulomb's Law - The magnitude $|\vec{\mathbf{F}}_\mathbf{E}|$ of the electrostatic force exerted by on point charge q_1 on another point charge q_2 is directly proportional to the magnitudes q_1 and q_2 of the charges and inversely proportional to the square of the distance r between them:

$$\vec{\mathbf{F}}_\mathbf{E} = \left(\frac{kq_1q_2}{r^2}\right)\hat{\mathbf{r}} \qquad |\vec{\mathbf{F}}_\mathbf{E}| = \frac{kq_1q_2}{r^2} \qquad \text{Units are Newton's (N)}$$

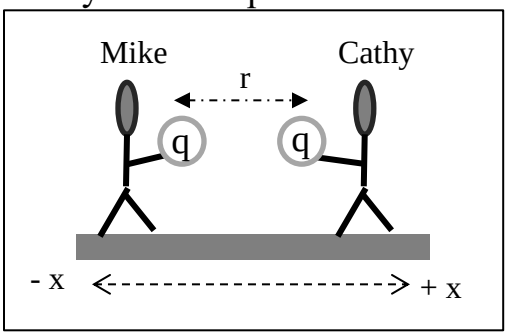
$\vec{\mathbf{F}}_\mathbf{E}$ – Electostaic force
$k = 1/4\pi\varepsilon_o$ where ε_o is the permittivity of free space ($\varepsilon_o = 8.85 \times 10^{-12}\ C^2/Nm^2$)
ε – The permittivity is a fundamental physical constant that quantifies how easily electric fields are established in the space around the charge.
$k = 9.0 \times 10^9\ Nm^2/C^2$ Electrostatic Constant for vacuum/air.
q_1 – net charge on object 1 – units are Coulombs.
q_2 – net charge on object 2 – units are Coulombs.
r – straight line distance between the two charged objects. (m)
$\hat{\mathbf{r}}$ – unit vector in the direction of the electrostatic force determined by the charge interaction.

- The direction of the electrostatic force is determined by the type of charges that are producing the force (-,- or +,+ or -,+) and is a vector quantity.
- The "sign" of the charge is not used in the force equation.
- The direction of the force determines the sign of the force and comes from the coordinate system.
- The force that q_1 experiences is equal in magnitude to the force q_2 experiences.
- This equation works for charged point sources and spherical bodies only.
- The force is the same for any point on the radius of the circle with radius "r". The distance from one charged object to the other is the radius of a circle and will produce the same magnitude of force at any point on this circle.



<u>Example</u>: Mike has a ball that has a net negative charge of 2 C and Cathy has a second ball with a net positive charge of ½ C,

A) How much force does Mike's ball exert on Cathy's ball if they are 1 m apart?



$k = 9.0 \times 10^9\ Nm^2/C^2$
$q_{Mike} = 2$ C negative charged
$q_{Cathy} = 0.5$ C positive charged
$r = 1$ m

$$\left|\vec{\mathbf{F}}_{\mathbf{E}}\right| = \frac{kq_1q_2}{r^2}$$

$$\left|\vec{\mathbf{F}}_{\mathbf{E}}\right| = \frac{(9.0\ x\ 10^9 Nm^2/C^2)(2\ C)(0.5\ C)}{(1m)^2} = 9.0\ x\ 10^9 N\ or\ about\ 2\ x\ 10^9 lbs$$

I don't think they would be holding on to the balls very long!

B) In what direction is the force exerted on Mike's ball?

Because the balls have opposite charge, they will attract therefore Mike's ball will experience an attractive force towards Cathy's ball in the positive x direction.

C) How much force does Cathy's ball exert on Mike's ball if they are 1 m apart?

$$\left|\vec{\mathbf{F}}_{\mathbf{E}}\right| = \frac{kq_1q_2}{r^2}$$

$$\left|\vec{\mathbf{F}}_{\mathbf{E}}\right| = \frac{(9.0\ x\ 10^9 Nm^2/C^2)(2\ C)(2\ C)}{(1m)^2} = 9.0\ x\ 10^9 N$$

Same magnitude of force that is exerted on Mike's

D) In what direction is the force exerted on Cathy's ball? Because the balls have opposite charge, they will attract therefore Cathy's ball will experience an attractive force towards Mike's ball in the negative x direction.

E) If Mike's ball has a mass of 2 kg, what is the ball's acceleration the instant Mike lets go of the ball? (ignore the gravitational force)

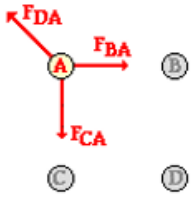
$\Sigma F_x = F_e = ma$ therefore $a = F_e/m = (9.0 \times 10^9\ N)/(2\ kg) = 4.5 \times 10^9\ m/s^2$

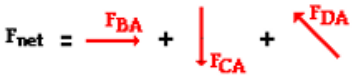
F) If both balls had either a positive or both balls had a negative charge what would this do to the direction of the force? The charge would be the same on both balls therefore the force would be repulsive. Mike's ball would be pushed in the negative x direction and Cathy's ball would be pushed in the positive x direction in either case, both positive charged or both negative charged.

Force due to more than two point charges - The net force acting on a charged object due to the electrostatic force is the vector sum of each individual force.

Superposition principle – The superposition principle states that when two or more physical influences (such as forces, waves, or quantum states) act on a system simultaneously, the resulting effect is the sum of the individual effects. For interacting forces adding individual force vectors will determine the magnitude and direction of the vector sum which is the net force or resultant force. The forces are superimposed on each other, and the resultant force is the net force the particle experiences.

$\Sigma\vec{F} = \vec{F}_{net} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \ldots$ vector sum



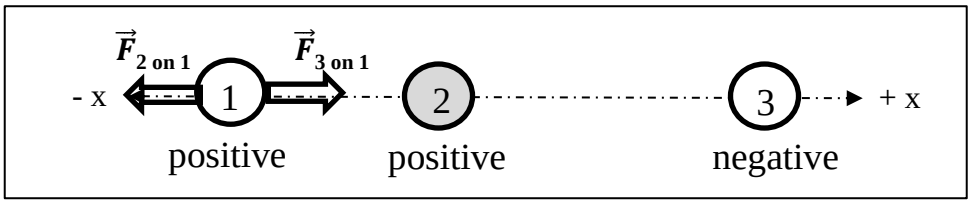


Net force due to multiple point charges (along a single axis) – if more than two point charges act on an object with a net charge the vector sum of the electrostatic force from each point source acting is the net force acting on the charged object. For the electrostatic force, the direction is taken into account by the "sign" of the force, which is obtained from the direction of the induvial forces and the defined directions of the x-axis (NOT the sign of the charge directly!)

<u>Example direction of forces</u>: If there are 3 balls with net charges as shown, what is the direction of the forces from each of the balls on each other?

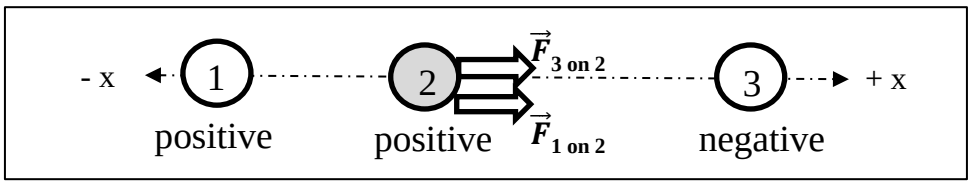
<u>Ball 1 -</u> There are two forces acting on Ball 1. There is the force that Ball 2 exerts on Ball 1, since both are positive Ball 2 will repel ball one in the negative x direction. The second force is from Ball 3 and since Ball 3 and Ball 1 are opposite charges Ball 3 will attract Ball 1 in the plus x direction. The net force on Ball 1 is

$$\Sigma F_{x\ Ball\ 1} = F_{3on1} + (-F_{2on1})$$



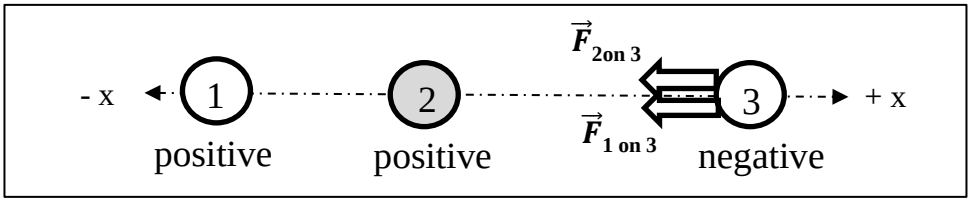
<u>Ball 2 -</u> There are two forces acting on Ball 2. There is the force that Ball 1 exerts on Ball 2, since both are positive Ball 2 will repel ball one in the positive x direction. The second force is from Ball 3 and since Ball 3 and Ball 2 are opposite charges Ball 3 will attract Ball 2 in the plus x direction. The net force on Ball 2 is

$$\Sigma F_{x\ Ball\ 2} = F_{3on2} + F_{1on2}$$

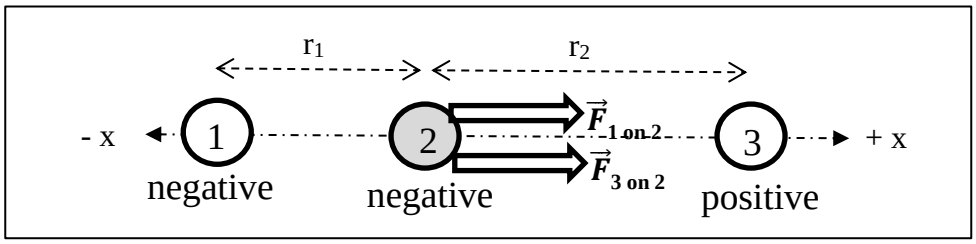


<u>Ball 3 -</u> There are two forces acting on Ball 3. There is the force that Ball 1 exerts on Ball 3, since they are opposite charge, Ball 1 will attract ball one in the negative x direction. The second force is from Ball 2 and since Ball 3 and Ball 2 are opposite charges Ball 2 will attract Ball 3 in the minus x direction. The net force on Ball 2 is

$$\Sigma F_{x\ Ball\ 3} = F_{1on3} + F_{2on3}$$



Example: If there are three balls with the net charge as shown, what is the direction of the net force on ball 2 (negative 4 μC), due to the charge on ball 1 (negative 2 μC) and ball 3 (positive 6 μC)? $r_1 = 0.1$ m and $r_2 = 0.2$ m, $k = 9.0 \times 10^9\ Nm^2/C^2$, 1 μC = 1×10^{-6} C. Assume the balls are not moving.



1) Only consider the magnitude and direction of the forces acting on ball 2. The forces on ball 2 are from the charge on ball 1 ($F_{1\text{ on }2}$) and the charge on ball 3 ($F_{3\text{ on }2}$), therefore there are two forces acting on ball 2.

2) Determine the direction of each of the forces by looking at the relative charge of the two objects.

   - Looking only at the force that ball 2 exerts on ball 1 ($\vec{F}_{1\text{ on }2}$), because they are the same charge ball 1 and 2 will repel each other, this makes the direction of the force on ball 2 due to ball 1 along the + x axis.
   - Looking only at the force that ball 2 exerts on ball 3 ($\vec{F}_{3\text{ on }2}$), because they are the opposite charge ball 3 will attract ball 2, this makes the direction of the force on ball 2 due to ball 2 along the + x axis.

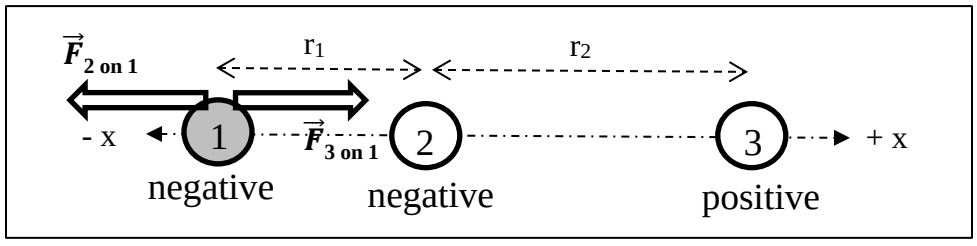
3) Calculate the magnitude of each of the forces using each pair's charge (q) and distance apart (r)

$$\left|\vec{F}_{1\text{ on }2}\right| = kq_1q_2/r_1^2 = (9.0 \times 10^9\ Nm^2/C^2)(2\times10^{-6}C)(4\times10^{-6}C)/(0.1m)^2 = 7.2\ N$$
$$\left|\vec{F}_{3\text{ on }2}\right| = kq_3q_2/r_2^2 = (9.0 \times 10^9\ Nm^2/C^2)(6\times10^{-6}C)(4\times10^{-6}C)/(0.2m)^2 = 5.4\ N$$

4) Sum the forces acting on the object in question and solve.

$$\Sigma F_{x\text{ on ball }2} = (+F_{1\text{ on }2}) + (+F_{3\text{ on }2}) = \text{net force on ball 2} = 7.2\ N + 5.4\ N = 12.6\ N$$

5) If the charge and distance stay the same on the balls, what is the magnitude and direction of the net force on ball 1?



The force that ball 1exerts on ball 2 is repulsive therefore is in the negative x direction. The force that ball 1 exerts on ball 3 is attractive therefore is in the positive x direction.

$$|\vec{\mathbf{F}}_{\mathbf{1\ on\ 2}}| = kq_1q_2/r_1^2 = 7.2\ \text{N}$$

The distance from ball 1 to ball 3 is $r_3 = r_1 + r_2 = 0.3$ m, this is the distance that is used in calculating the force from ball 3 on ball 1.

$$|\vec{\mathbf{F}}_{\mathbf{3\ on\ 1}}| = kq_3q_1/r_3^2 = 1.2\ \text{N}$$

$$\Sigma F_{x\ \text{on ball 2}} = (-\mathbf{F_{2\ on\ 1}}) + (+\mathbf{F_{3\ on\ 1}}) = \mathbf{-\ 7.2\ N + 1.2\ N = -\ 6.0\ N}$$

Net force due to multiple point charges (in a plane) – If the charged objects are not along the same line but are in 2-dimensional space then the magnitude and direction of the electrostatic forces must be taken into account by finding the resultant for the forces acting.

$$\mathbf{\Sigma\vec{F}} = (\Sigma F_x)\hat{\mathbf{x}} + (\Sigma F_y)\hat{\mathbf{y}} = m\vec{\mathbf{a}}$$

If there are two Electric forces given by

$$\vec{\mathbf{F}}_{\mathbf{1}} = (F_{1x})\hat{\mathbf{x}} + (F_{1y})\hat{\mathbf{y}} \qquad \vec{\mathbf{F}}_{\mathbf{2}} = (F_{2x})\hat{\mathbf{x}} + (F_{2y})\hat{\mathbf{y}}$$

Taking the dot product with the unit vector $\hat{x}$ to find the net force in the x-direction

$$\mathbf{\Sigma\vec{F}} \bullet \hat{\mathbf{x}} = \left[(\Sigma F_x)\hat{\mathbf{x}} + (\Sigma F_y)\hat{\mathbf{y}} + (\Sigma F_z)\hat{\mathbf{z}}\right] \bullet \hat{\mathbf{x}} = [m\vec{\mathbf{a}}] \bullet \hat{\mathbf{x}}$$

$$\Sigma F_x = F_{1x} + F_{2x} = ma_x$$

Taking the dot product with the unit vector $\hat{y}$ to find the net force in the y-direction

$$\mathbf{\Sigma\vec{F}} \bullet \hat{\mathbf{y}} = \left[(\Sigma F_x)\hat{\mathbf{x}} + (\Sigma F_y)\hat{\mathbf{y}} + (\Sigma F_z)\hat{\mathbf{z}}\right] \bullet \hat{\mathbf{y}} = [m\vec{\mathbf{a}}] \bullet \hat{\mathbf{y}}$$

$$\Sigma F_y = F_{1y} + F_{2y} = ma_y$$

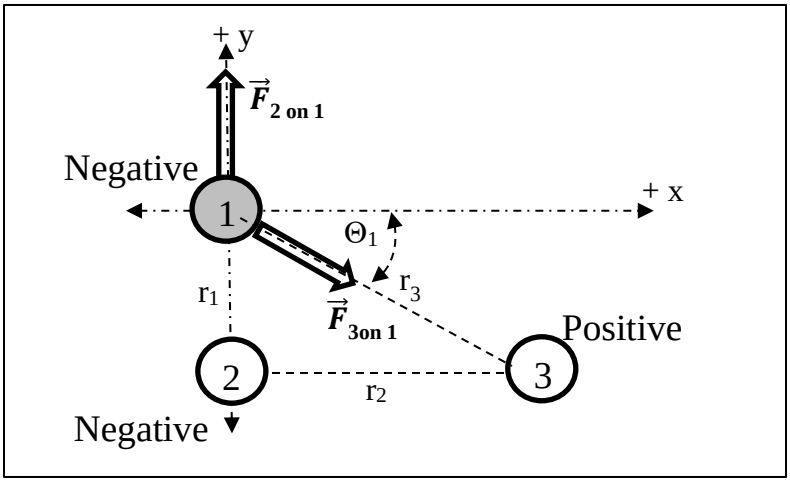
The net force is then equal to

$$\mathbf{\Sigma\vec{F}} = (F_{1x} + F_{2x})\hat{\mathbf{x}} + (F_{1y} + F_{2y})\hat{\mathbf{y}} = m\vec{\mathbf{a}}$$

- ✓ Use any symmetries of the charge distribution to make the problem easier!

Example: There are three balls with the net charges as shown, what is the magnitude and direction of the net force on ball 1 (2 µC, negative charge), due to the charge on ball 2 (4 µC, negative charge) and ball 3 (6 µC, positive charge)? k = 9.0 x 10^9 Nm^2/C^2 (Assume the balls are not moving)

r_1 = 0.1 m and r_2 = 0.2 m, r_3 = 0.224 m, $\Theta_1 = 26.6°$



1) Only consider the forces (magnitude and direction) acting on ball 1. The forces that are exerted on ball 1 are from the net charge on ball 2 ($\vec{F}_{2\text{ on }1}$) and the net charge on ball 3 ($\vec{F}_{3\text{ on }1}$), there are the two forces acting on ball 1.

2) Determine the direction of each of the forces by looking at the relative charge of the two objects.

   - Looking only at the force that ball 1 exerts on ball 2 ($\vec{F}_{2\text{ on }1}$), because they are the same type of charge ball 1 and 2 will repel each other, this makes the direction of the force on ball 2 due to ball 1 along the + y axis.
   - Looking only at the force that ball 3 exerts on ball 1 ($\vec{F}_{3\text{ on }1}$), because they are the opposite type of charge ball 3 will attract ball 1, this makes the direction of the force on ball 1 due to ball 2 at $\Theta_1 = 26.6°$ below the + x axis.

3) Calculate the magnitude of each of the forces using each pair's charge (q) and distance apart (r)

$$\left|\vec{\mathbf{F}}_{\mathbf{2\text{ on }1}}\right| = \frac{kq_2q_1}{r_1^2} = \frac{(9.0\text{ x }10^9\text{ N}m^2/C^2)(2\text{x}10^{-6}\text{C})(4\text{x}10^{-6}\text{C})}{(0.1\text{m})^2} = 7.2\ N$$

$$\left|\vec{\mathbf{F}}_{\mathbf{3\text{ on }1}}\right| = \frac{kq_3q_1}{r_3^2} = \frac{(9.0\text{ x }10^9\text{ N}m^2/C^2)(6\text{x}10^{-6}\text{C})(2\text{x}10^{-6}\text{C})}{(0.224\text{m})^2} = 2.15\ N$$

Note: The calculated magnitude of the forces does not have a sign (+ or -)!

4) Find components of the force vectors and put in component vector form.

$\vec{\mathbf{F}}_2 \bullet \hat{\mathbf{x}} = F_{2x} = \left|\vec{\mathbf{F}}_2\right|\cos(90°) = 0$ N
$\vec{\mathbf{F}}_2 \bullet \hat{\mathbf{y}} = F_{2y} = \left|\vec{\mathbf{F}}_2\right|\cos(0°) =$ + 7.2 N (along + y axis)
$\vec{\mathbf{F}}_3 \bullet \hat{\mathbf{x}} = F_{3x} = \left|\vec{\mathbf{F}}_3\right|\cos(26.6°) =$ + 1.92 N (x component along + x-axis)
$\vec{\mathbf{F}}_3 \bullet \hat{\mathbf{y}} = F_{3y} = \left|\vec{\mathbf{F}}_3\right|\cos(90° + 26.6°) =$ - 0.96 N (y component along – y-axis)

$\vec{F}_2 = (0\text{ N})\hat{x} + (7.2\text{ N})\hat{y}$ and $\vec{F}_3 = (1.92\text{ N})\hat{x} + (-0.96\text{ N})\hat{y}$

5) Sum the forces acting on the object in question and solve.

$$\Sigma\vec{\mathbf{F}} = (F_{1x} + F_{2x})\hat{\mathbf{x}} + (F_{1y} + F_{2y})\hat{\mathbf{y}}$$

$$\Sigma\vec{\mathbf{F}} = (0\text{ N} + 1.92\text{ N})\hat{\mathbf{x}} + (7.2\text{ N} + (-0.96\ N))\hat{\mathbf{y}}$$

The net force in components form is given as

$$\Sigma\vec{\mathbf{F}} = (1.92\text{ N})\hat{\mathbf{x}} + (6.24\ N)\hat{\mathbf{y}}$$

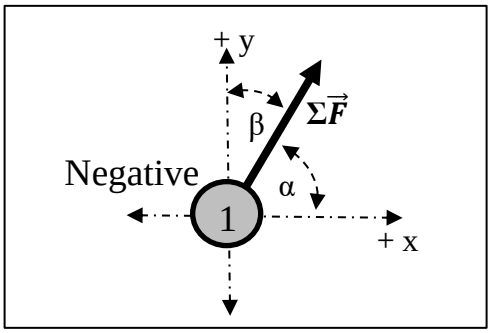
The magnitude of the net force is given by

$$\left|\Sigma\vec{F}\right| = \sqrt{\Sigma F_x^2 + \Sigma F_y^2} = 6.5\ N$$

The direction of the net force is given by

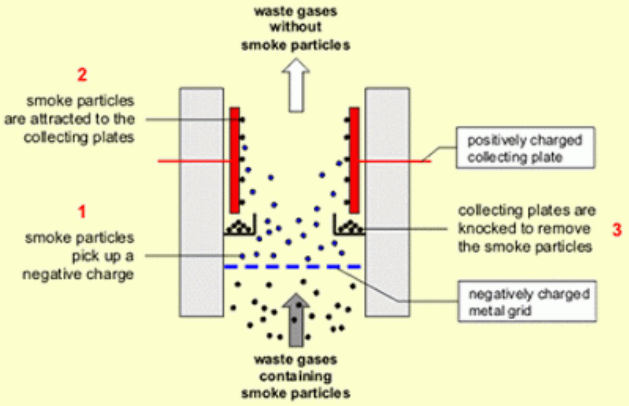
$$\alpha_{net} = cos^{-1}\left(\frac{\Sigma F_x}{\left|\Sigma\vec{F}\right|}\right) = 73° \qquad \beta_{net} = cos^{-1}\left(\frac{\Sigma F_y}{\left|\Sigma\vec{F}\right|}\right) = 17°$$

Therefore ball 1 experiences a force with a magnitude of 6.5 N at an angle of 73° measured from the positive x-axis due to the charge on ball 1 and 2.

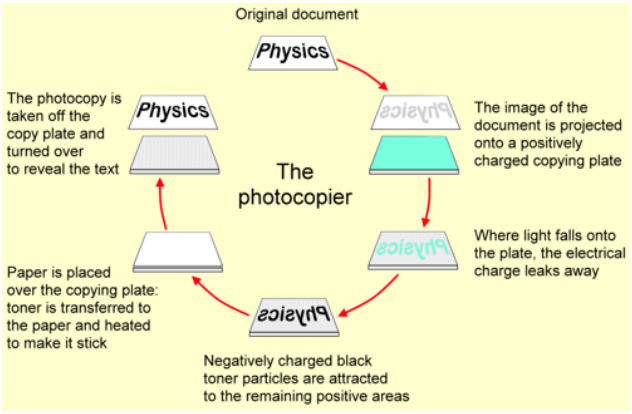


Examples and applications of Electrostatic Force

- Electrostatic Filters and smoke detectors – use an electric field to ionizes (charge) dust or smoke particles and attracts them to a positive plate.



- Copiers – High speed copiers use light (image of what is to be copied) to remove charge where the light shines on the copy plate (no type or image) and does not remove as much charge where it is dark (type or image), when the toner is released it "sticks' to the positive charged plate and the negative charged toner sticks to the positive charge on the plate. The piece of paper is placed on the plate and the toner is transferred to the paper; the paper is then heated to fix the toner to the paper.



- Painting a car – uses charge to make paint stick better and more efficently to postively charged car body and negatively charged paint.

